

STREAMLIT DASHBOARD REPORT

Product Line Profitability and Margin Performance Analysis Using Streamlit

Internship Project Report

Submitted to: Unified Mentor Pvt. Ltd.

Submitted by: Tina Kumbhare

Course: Data analyst

Tools Used

Python

Streamlit

Pandas

Plotly

Power BI

Visual Studio Code

Abstract

This project presents an interactive Streamlit dashboard developed for analyzing the profitability and margin performance of products using the Nassau Candy Distributor dataset. The dashboard enables users to monitor key business metrics such as Total Sales, Total Cost, Total Profit, and Profit Margin in real time. Interactive filters and dynamic visualizations provide business users with an intuitive platform for exploring product-level and division-level performance. The dashboard simplifies decision-making by transforming raw transactional data into meaningful insights and supports business intelligence through an accessible web interface.

1. Introduction

Business Intelligence dashboards help organizations convert complex datasets into visual reports that support quick and informed decision-making. While Power BI provides rich analytical capabilities, Streamlit offers a lightweight and interactive platform for deploying dashboards as web applications using Python.

The objective of this project is to demonstrate how the Nassau Candy Distributor dataset can be analyzed through a Streamlit application, allowing users to interactively explore product profitability, compare divisions, and monitor key performance indicators.

2. Project Objectives

The main objectives are:

- Develop an interactive web dashboard using Streamlit.
- Display business KPIs in real time.
- Enable dynamic filtering of products and divisions.
- Visualize profitability using interactive charts.
- Improve business decision-making through easy-to-use analytics.
- Support future deployment on cloud platforms.

3. Technologies Used

Technology	Purpose
Python :-	Data processing
Streamlit :-	Web dashboard development
Pandas :-	Data manipulation
Plotly :-	Interactive charts
PowerBI:-	Dashboard design and validation
Visual Studio:-	Development environment
Code	

4. Dataset Description

The dashboard uses the Nassau Candy Distributor dataset containing product sales transactions. The dataset includes product details, division, sales, cost, profit, customer information, dates, and

regional data.

The application processes the dataset using Pandas before generating interactive visualizations.

5. System Architecture

The Streamlit dashboard follows a simple workflow:

Nassau Candy Distributor Dataset (CSV)



Data Cleaning (Pandas)



KPI Calculation & Aggregation



Interactive Charts (Plotly)



Streamlit Dashboard Interface



Business Users / Managers

This architecture ensures efficient processing and presentation of business data.

6. Dashboard Features

The Streamlit dashboard includes:

KPI Cards

- Total Sales
- Total Cost
- Total Profit

- Profit Margin

Interactive Filters

- Product Name
- Division
- Date Range
- Margin Threshold

Visualizations

- Product Profitability Leaderboard
- Profit Contribution Chart
- Revenue vs Profit Comparison
- Margin Distribution
- Cost vs Sales Scatter Plot
- Pareto Chart
- Margin Risk Table

6. Dashboard Workflow

1. Load CSV dataset.
2. Clean and preprocess data.
3. Calculate KPIs.
4. Apply user-selected filters.
5. Generate interactive charts.
6. Display dashboard in Streamlit.
7. Update visualizations dynamically based on user inputs.

7. User Interface

The dashboard provides:

- Clean and responsive layout.
- Sidebar filters.
- KPI cards at the top.
- Interactive Plotly charts.
- Download option for filtered data.

- Responsive design for desktop and laptop users.

8. Advantages

Interactive analytics.

- Fast performance.
- Easy deployment.
- User-friendly interface.
- Supports real-time decision-making.
- Reduces manual reporting effort.

9. Future Enhancements

Future improvements include:

- Live database connectivity.
- Authentication and user roles.
- Predictive analytics using machine learning.
- Sales forecasting.
- Inventory optimization.
- Cloud deployment using Streamlit Community Cloud.

10. Conclusion

The Streamlit dashboard enhances business intelligence by providing an interactive and efficient platform for profitability analysis. It complements the Power BI dashboard by enabling web-based analytics with Python, allowing users to explore sales performance, profitability, and operational metrics in real time. The solution supports data-driven decision-making and offers scalability for future enhancements.